

JERICO B. BAAL

ELECTRONICS ENGINEER & TECHNICIAN

Design Engineer | Special Science Teacher | Embedded Systems



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EDUCATION

Bachelor of Science in Electronics and Communications Engineering | 2015 - 2020

- Polytechnic University of the Philippines - Sta. Mesa, Manila
- President's List A.Y. 2019-2020
- Department of Science and Technology (DOST - SEI) - R.A. 10612 Scholar, 2017 - 2020

LICENCE AND CERTIFICATIONS

- Electronics Engineer 0078418 - Professional Regulatory Commission (2022)
- Electronics Technician 0025290 - Professional Regulatory Commission (2022)
- Data Science Essentials With Python - A Cisco Networking Academy program by DICT (2025)
- Full Stack Web Development Bootcamp - UDEMY , Dr. Angela Yu

PROFESSIONAL EXPERIENCE

Special Science Teacher I - DepEd Bacoar City | October 2022 - Present

- Robotics training covering Arduino/ESP32 C programming, sensor interfacing, GPIO control
- Engineered competitive robotics systems from concept to deployment for National Competitions
- Developed student IoT research projects and simulation research using modeling tools
- Teach various scientific disciplines including Life Science, Chemistry, Physics, and Earth Science

Design Engineer - DENSO Philippines | January 2021 - September 2022

- Validated vehicular smart key system designs based on engineering drawing specifications
- Conducted laboratory tests on RF parameters (frequency, amplitude) using anechoic chamber
- Test for board-level functionality, including voltage and current consumption and EEPROM
- Implemented testing protocols ensuring compliance with regulatory and quality standards
- Used Siemens NX and ELSA schematic tools for engineering drawing and schematic analysis
- Performed benchmarking and monthly technical reporting for new RF electronic products

TECHNICAL SKILLS

- Embedded Systems - Arduino, ESP32, peripherals (GPIO, ADC), modules, IoT development
- Programming - C, C++ Python, VS Code, STM32 Cube IDE
- Electronics and Hardware Design - Schematic & PCB Design (KiCad), circuit debugging and testing
- Engineering & Lab Tools - Siemens NX , Git (version control), anechoic chamber
- Web Development - HTML, CSS, Javascript, Node.js, Express.js, React.js, PostgreSQL

ACHIEVEMENTS & PROJECTS

- **1st Place, 1kg and 3kg Sumobot RC - National Science Quest 2025** - Association of Science Educators in the Philippines, Teacher's Camp, Baguio City
- **Qualifier, Regional Science and Technology Fair 2024 (CALABARZON)** - Robotics and Intelligent Machines - "PROJECT SAFE: SMS -Based Alerts, Air Quality Monitoring, Fire Surveillance and Environmental Sensing through the utilization of ESP32 Platform"
- **2nd Place, Division Science and Technology 2025 Fair (Bacoar City)** - Mathematics and Computational Sciences - Comparative SUMO Simulation of Default and "Buhos" Traffic Control Schemes Across Varying Demand Levels at Selected Junctions and Intersections in Bacoar City, Cavite: Traffic Performance Analysis"

Character reference is available upon request